

Veena S. Chakravarthi

A Practical Approach to VLSI System on Chip (SoC) Design

A Comprehensive Guide



Springer

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A comprehensive overview of the design criteria, methodology, skills, and knowledge needed for an SOC VLSI designer. It enables fresh engineering graduates to contribute in the industry from day one and create complex SOC designs

Veena S. Chakravarthi

Dedicated to VLSI designers

Foreword

It's an excellent time to be working in the semiconductor industry. Qualitatively, we are all familiar with Generation Z's constant appetite for digital consumption. That appetite is driving technical innovation starting in huge data centers and moving out to the growing sea of smartphones. Quantitatively, Gartner tells us that our industry is growing at a rate of 26% year over year. The semiconductor industry has never been more complex, and it's going to keep getting more complicated. Every device needs to be smaller, more powerful, and more energy-efficient than the previous generation.

There is no doubt our industry is shifting as waves of consolidation and innovation crash into new geographies and new markets, but the demand for intelligent, highly integrated chip design keeps growing. This means that any aspiring hardware engineer – whether they want to work for a hungry, young startup or an established house of silicon – needs to become fully versed in the art of very large-scale integration (VLSI). There is no better teacher to learn from than Dr. Veena Chakravarthi.

I first met Veena in 2003 when she joined Centillium to play a key role in developing the high-performance system on chip (SOC) solutions for Ethernet Passive Optical Networks (EPON). Those products helped us enable Asian service providers to deliver some of the first fiber to the home deployments in the world and threw fuel on the fire of data consumption. I've followed her career ever since as she continues to add technical, professional, and academic accolades to a stellar resume.

With 30 years of experience as an SoC architect and VLSI designer, Veena has distinguished herself as both an artist and an engineer. Her abilities to design large, complex electronic systems in silicon have created baseline, enabling technologies for a number of communications systems. Her depth of experience has allowed her to create a perfect primer for any engineer wanting to arm themselves with the necessary mindset to understand the chip design process and development cycle for SoCs. This practical approach contains straight forward applications of known techniques to create a structure which will help freshman engineers contribute effectively to the SoC design and development process.

I'm excited about the future of our industry and where SoCs can take us. They are at the heart of the advancements in medical, biotech, transportation,

telecommunication, and countless other industries that will change how we live. This book is a thoughtful guide for any aspiring chip designer, and I thank Veena for teaching the next generation of innovators, inventors, and dreamers.



CEO and Chairman, Aquantia Corporation
San Jose, CA, USA

Faraj Aalaei

Foreword

The semiconductor industry is undergoing a massive change with technologies like IOT, intelligent edge/cloud, mobility, automotive, 5G, AI, and ML, creating in major opportunities. The expectations of 50 billion connected devices by 2025 and the massive amounts of data that will need to be processed on edge analytics as well as on cloud will result in sharper insights for better decision-making.

With customers expecting continual improvements in applications, the question is whether the chip industry is moving fast enough to meet these expectations. A broad supply chain, equipment, and materials innovations and attracting the “best of the best” college graduates to fuel innovation are key.

This is an excellent time for young engineers to make the most of the opportunities and thereby fulfil their career aspirations, be it in corporate or entrepreneurship. The book *A Practical Approach to VLSI System on Chip (SoC) Design* by Prof. Veena Chakravarthi is a good reference guide for new engineers and also a good refresher for seasoned practitioners of VLSI.

I have known Veena since early 2000 when she joined the core team of the technology business at Mindtree when she played a crucial part in developing successful in-house IPs like Bluetooth and WLAN core. She is a seasoned designer as well as an academician. Her experiences would be useful for both industry and academic needs and help engineers to take up path breaking design challenges.



Executive Chairman, Happiest Minds
Bangalore, India

Ashok Soota

Foreword

VLSI design of “systems on chip” (SoCs) has suddenly taken a change in direction. Traditional computer architectures can no longer solve the computing problems of tomorrow. New, innovative approaches to SoC design will use non-Von Neumann architectural approaches with embedded neural networks to make problems like pattern recognition solvable in real time. Suddenly, the world of venture capital-funded fabless semiconductor companies has exploded, as these companies propose innovative SoCs to solve “domain-specific” problems like vision-, sound-, or smell-related pattern recognition. Being able to do a few specific types of operations extremely well now becomes much more important than doing a wide variety of things very well. Beginning in the second half of 2017, the amount of venture capital money invested in fabless semiconductor and IP startups has accelerated, reaching an all-time record in 2018.

Books like *A Practical Approach to VLSI System on Chip (SoC) Design* provide guidance for aspiring designers and academics who wish to join this parade of innovation. Rarely do opportunities like this emerge in the semiconductor industry. But this is a time of new ideas where the ability to translate algorithmic innovation to silicon can drive quantum steps forward in machine learning capability. The first wave of semiconductor technology was driven by physical component innovation. This wave will be driven by system innovation, combining unique software with clever hardware architectures. It will be an exciting revolution in computing.



CEO Emeritus of Mentor Graphics, A Siemens Business
Dallas, US

Walden C. Rhines

Preface

Having worked in semiconductor design industry for over two decades, it was my strong desire to pass on the knowledge of system on chip design to the next generation. Therefore, I conceived the idea of writing a book on “A Practical Approach to VLSI System on Chip (SoC) Design.”

The book intends to present a comprehensive overview of the design methodology, environment, and requisite skills that are required for design and development of system on chip (SOC).

It ensures that engineers are aware and are able to contribute effectively in fab-less design companies from day one up to the development of complex SOC designs.

While this book is targeted for electrical and electronic engineers who aspire to be VLSI designers, it is also a valuable reference guide for professional designers who are part of the development teams in VLSI design centers – the ones behind complex systems on chip solutions.

The book aims to give the readers a comprehensive idea of what one has to do as a VLSI designer. It expands on the arsenal of skills they need to be equipped with, the responsibilities of the job, and the challenges that they should anticipate. This information is based on my experiences in the semiconductor industry and academics since the past 25 years.

Typically, electronic engineers aspire to become VLSI designers either during or after their undergraduate or graduate studies. Unfortunately for them, they usually don't possess the requisite skills and design techniques to circumnavigate the challenges they'll face in the industry. Meanwhile, young VLSI designers in the industry struggle to see the big picture of the design process. It's not practical for one person to work in all areas of the VLSI design and development process. This book is my attempt to provide answers to both groups, so that they can plan, understand, and equip themselves with necessary skill sets. The design case relevance in every chapter and the design examples in Chap. 11 help the readers realistically visualize problems and solutions encountered during VLSI system design.

The target audience for this book are engineering students who are pursuing a degree in Electrical, Electronics, and Communication and allied branches like Biomedical, Biotechnology, Instrumentation, Telecommunication, etc. Also,

engineers in early stages of their career in the semiconductor industry can refer to the book for a complete understanding of the chip design process.

Though, the book covers the complete spectrum of the topics relevant to system on chip (SoC) using VLSI technology, it is good to have a fundamental understanding of the logic design as it is a prerequisite to follow the contents of the book.

Though India is seen as silicon country with Bangalore as silicon city with many fabless design centers in VLSI, it is facing acute shortage of employable VLSI design engineers as large number of fresh engineers graduating from universities are not readily deployable for design jobs.

Statistics show that there is a demand of over 3000 design engineers per annum and will soon grow up to 30,000 per annum in the coming years. Engineering schools currently are catering to only 50% of the annual demand. Globally, the scenario is not too different.

In this scenario of shortage, a VLSI design engineer has a promising and bright future ahead and can expect a challenging and rewarding career. Globally, the semiconductor industry is one of the fastest-growing industries at 26% annually according to Gartner's recent market research and so are VLSI design jobs. Skilled VLSI persons are always in demand in catering the most challenging system on chip designs, the new versions of EDA tools addressing heterogeneous complex system integrations, the fabrication technology correlations, etc. Countries like Egypt need around 10,000 skilled VLSI designers.

The design productivity gap – a shortage of skilled manpower that can convert transistors (that fabrication technology offers) to useful ones – is real. Hence, there is a need to develop skill sets to suit the semiconductor jobs and bridge this gap.

It would not have been possible to realize this project without the support of many of my friends, colleagues and family. First, I wish to thank my father, Mr. R S Chakravarthi, a noted journalist and a Rajyotsava awardee from Karnataka, India, whose literary gene was responsible for harboring my desire to write a book. My heartfelt thanks to my loving family, my husband, Dr. K S Sridhar, and sons, K S Abhinandan and K S Anirudh. I am indebted to my colleague, Dr. M S Suresh, Scientist, ISRO, who patiently read each of my chapters and offered line-by-line reviews.

I wish to thank my ex-colleagues Mr. Sathish Burli for describing the software development flow, Dr. K S R C Murthy for sharing information on packaging with me, and Mr. Dinesh for identifying IOT-SOC reference design which is available in www.opencores.org for the case study. My steadfast team, comprising of Vaibhav, Om Prakash, and my dear students Amruthashree and Aditya, tried out all the design examples and ensured that they are working and ready for the reference. Thanks to them.

I'm also grateful to the semiconductor industry for having embraced me so warmly. And I'm mighty thankful to Mr. Faraj Aalaei, executive CEO, Aquantia Inc.; Mr. Ashok Soota, executive CEO, Happiest Minds; and Mr. Walden C Rhines, emeritus CEO, Mentor Graphics, Siemens group, for taking time out of their busy schedules to write the foreword for this book.

I thank all the organizations I have worked with for contributing directly or indirectly to the naming of this book. Special thanks go to BNMIT for encouraging me to pursue this endeavor.

Last but not the least, I thank my super power who gives me the motivation and constant energy to take up projects beyond my capability and make it happen.

I will be very happy if the users find each chapter useful and try out design examples and reference design and subsequently make VLSI their career choice. I am curious about your feedback and criticisms. I'm sure it'll go a long way in bettering this book.

Thank you.



System on Chip Architect, Bangalore, India

Veena S. Chakravarthi

Why This Book?

Why One Should Read This Book?

This book is intended for the electrical and electronics graduate and undergraduate students of engineering schools who aspire to be VLSI designers. It can also be referred by the engineers and professional designers who are part of the development teams in VLSI design centers. It aims to give the readers complete perspective of what one has to do as a VLSI designer and the skill set required for them, the job content, and the challenges faced. The information is based on the personal experience the authors have in their semiconductor industry and academic career spreading over two and half decades.

What Problem Does It Solve?

Typically, the electronic engineers during their undergraduate and graduate courses aspire to become VLSI designers but would not know what necessary skill set to possess, job content, design techniques, and the challenges they get to face. Paradoxically, VLSI designer in the industry will not have a big picture of the design process as it is not practical for anyone to work in all areas of the VLSI design and development process. This book attempts to provide answers to both of them, so that they can plan, understand, and equip themselves with necessary skill sets. The design scenarios, in every chapter, helps one to visualize the problems and the solutions encountered during the VLSI system design realistically.

Who Are the Audience?

Engineering students with Electrical, Electronics, and Communication and allied branches like Biomedical, Biotechnology, Instrumentation, Telecommunication, etc. aspiring to be VLSI designers can follow this as guide to understand and learn the skill set required to become VLSI designers. Also, engineers in early stage of career who have joined companies in semiconductor industry can refer to the book for the complete understanding of chip design process and relate their work to get the complete process of the design and development cycle of the system on chip.

What Are the Prerequisites to Read This Book?

Though the book covers complete spectrum of the topics relevant to system on chip (SoC) using VLSI technology, it is good to have a fundamental understanding of the logic design as the pre-requisite to follow the contents of the book. The book is targeted to undergraduate and graduate students of Electrical and Electronics Engineering and allied courses which have logic design as a subject.

Why Become VLSI Designer?

Though India is seen as silicon country with Bangalore as silicon city with many fabless design centers in VLSI, it is facing acute shortage of employable VLSI design engineers as large number of fresh engineers graduating from universities are not readily deployable to the design jobs. Statistics shows that there is a demand of over 3000 design engineers per annum and will soon grow up to 30,000 per annum in the coming years. The engineering schools are currently catering to only 50% of the demand annually. Globally, the scenario is not different. In this scenario of shortage, a VLSI design engineer has a promising and bright career prospects, with a challenging and a technically satisfying career.

Globally, the semiconductor industry is one of the fastest growing at 16% annually according to Gartner's recent market research [1] and so are VLSI design jobs. Skilled VLSI persons are required to cater the most challenging system on chip designs, the new versions of EDA tools addressing heterogeneous complex system integrations, the fabrication technology correlations, etc. Countries like India need around 3000 skilled VLSI designers for around 150 companies working in design space as reported in the 28th International Conference on VLSI Design held in Bangalore, India. That means design productivity gap – the shortage of skilled manpower who can convert the number of transistors the fabrication technology offers to functionally useful ones – exists. Hence, there is a need to develop a skill set to suit the semiconductor jobs that will help in bridging this gap.

How Is the Book Organized?

The book chapters primarily target digital SOC with few analog/mixed signal blocks by addressing their integration to digital SOC. At the end of the book, the reader should get the fair idea of SOC by definition, constituents and their selection, parallel design and integration flows, design infrastructure needs, skill set required, automated design flows like synthesis, physical design, design for testability, static timing analysis, and packaging. Detailed explanation of any of these processes is not the intent of the book; however, it is aimed to cover the entire design process from the specification to tapeout and introduction to packaging. The design examples given in Chap. 12 are small functional blocks with the testbench and reference waveform, which should bring up the reader to try hands-on design process. However, it requires the EDA tools and the standard cell library to carry out the design. The design cases give practically fair idea of how the design blocks of medium complexity is done which can be further extended to the design of SOC. Book organization is as follows:

Chapter 1 introduces the SOC trends in terms of complexity, die size, speed of operation, and drivers of the phenomenal advancement in VLSI. It lists some of the major challenges of SOC design.

Chapter 2 explains the SOC design and the design flow.

Chapter 3 deals with the constituents of SOC and the selection criteria of each of them.

Chapter 4 details the design process by standard industry followed method for modelling using HDL – Verilog.

Chapter 5 explains the process of SOC synthesis.

Chapter 6 explains the static timing analysis, STA.

Chapter 7 deals with the design for testability of SOC.

Chapter 8 deals in detail the need for verification, Verification methods and related processes like coverage, Bug tracking, sanity and regression and formal verification.

Chapter 9 explains the physical design of the SOC and few advanced techniques being followed for low power, advanced technology, and preferred data path SOC.

Chapter 10 deals with the physical design verification procedures for SOC design.

Chapter 11 introduces packaging technology and options available for SOC.

Chapter 12 has a set of design examples, design flow, and reference to case study to try hands-on.

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Abbreviations and Acronyms

ADC	Analog to Digital Converter
AHB	Advanced High-Performance Bus
AMP	Asymmetric Multiprocessing
API	Application Program Interface
ASIC	Application-Specific Integrated Circuit
ASCII	American Standard Code for Information Interchange
ATE	Automatic Test Equipment
ATPG	Automatic Test Pattern Generation
ATSE	Advanced Television Systems Committee
BCL	Base Class Library
BGA	Ball Grid Array
Bi-CMOS	Bipolar Complementary Metal-Oxide Semiconductor
BIST	Built-In Self-Test
BS	Boundary Scan
BFM	Bus Functional Model
CIF	Caltech Intermediate Format
CMOS	Complementary Metal-Oxide Semiconductor
CSP	Chip-Scale Packaging
CTS	Clock Tree Synthesis
CVD	Chemical Vapor Deposition
DAC	Digital to Analog Converter
DDR	Double Data Rate
DEF	Design Exchange Format
DFT	Design for Testability
DMAC	Direct Memory Access Controller
DRC	Design Rule Check
DRM	Design Rule Management
DRV	Design Rule Violation
DUT	Design Under Test
ECO	Electronics Change Order
EDA	Electronic Design Automation

EM	Electromigration
ERC	Electric Rule Check
ESD	Electrostatic Discharge
EU	Effective Utilization
EVM	Electronics Validation Module
Fab-less	Companies which do all services except the wafer and chip fabrication process
FCS	Frame Check Sequence
FBGA	Fine Pitch Ball Grid Array
FET	Field Effect Transistor
FPGA	Field Programmable Gate Array
FPU	Floating Point Unit
FSM	Finite-State Machine
FIFO	First In First Out
FTP	File Transfer Protocol
GALS	Globally Asynchronous Locally Synchronous
GDS II stream format	Graphic database system II stream format, an industry standard format in which the IC design layout with name convention is represented
GSLA	Globally Synchronous and Locally Asynchronous
HDL	Hardware Description Language
HFN	High Fanout Nets
HLD	High-Level Design Document
IC	Integrated Circuit
IEEE-SA	Institute of Electrical and Electronics Engineers Standards Association
I2C	Inter-integrated Circuit
ICG	Integrated Clock Gate
I2R	Input to Register
I2O	Input to Output
IO	Input-Output
IP Cores	Intellectual Property Cores
ISP	In-System Programming
ITU-T	International Telecommunication Union-Telecommunication
JTAG	Joint Test Action Group
LAN	Local Area Network
LBIST	Logic Built-In Self-Test
LC	Inductance-Capacitance
LEC	Logic Equivalence Check
LEF	Library Exchange Format
LFSR	Linear Feedback Shift Register
LIB	Liberty File Format
LINT	Tool that analyze programming and flag errors based on set of rules defined

LVS	Layout Versus Schematic
MBIST	Memory Built-In Self-Test
MCM	Multi-chip Module
MIL	Military
MIPS	Million Instructions per Second
MRD	Market Requirement Document
MISG	Multiple Input Sequence Generator
MEMs	Microelectromechanical Systems
MoCA	Multimedia over Coax Alliance
MSV	Multiple supply voltage
MSSV	Multi-supply Single Voltage
NAS	Network-Attached Storage
NRE	Nonrecurring Engineering
OCV	On-Chip Variation
OS	Operating System
OSCG	On-SOC Clock Generation
PCB	Printed Circuit Board
PGA	Pin Grid Array
P&R	Place and Route
PR Boundary	Place and Route Boundary
PRD	Product Requirement Document
PRPG	Pseudorandom Pattern Generator
PTAM	Power-Aware Test Access Mechanism
PLL	Phase-Locked Loop
PMBIST	Programmable Memory Built-In Self-Test
PVD	Physical Vapor Deposition
PVT	Process-Voltage-Temperature
RC	Resistance-Capacitance
RTL	Register Transfer Level
ROI	Return on Investment
R2R	Register to Register
R2O	Register to Output
SAN	Storage Area Network
SEM	Scanning Electron Microscope
SDC	Synthesis Design Constraint
PDP	Preferred Data Path
SDF	Standard Delay Format or Synopsys Delay Format
SI	Signal Integrity
SIP	System in Package
SLEC	Sequential Logic Equivalence Check
SMD	Surface Mount Device
SSN	Simultaneous Switching Noise
SPEF	Standard Parasitic Exchange Format
SPI	Serial Peripheral Interface
SPICE	Simulation Program with Integrated Circuit Emphasis

SMP	Symmetric Multiprocessing
SOC	System on Chip
SRAM	Static Random-Access Memory
STA	Static Timing Analysis
STUMP	Self-Test Using MISR and Parallel SRPG
TLF	Timing Liberty Format
TPI	Test Program Interface
TSMC	Taiwan Semiconductor Manufacturing Company
QFP	Quad Flat Package
UART	Universal Asynchronous Receiver-Transmitter
USB	Universal Serial Bus
UV	Ultraviolet
UVM	Universal Verification Methodology
VHDL	VLSI Hardware Description Language
VIP	Verification Intellectual Property
VLSI	Very Large-Scale Integration
WIFI	Wireless Fidelity
WSP	Wafer Scale Packaging
WNS	Worst Negative Slack

Chapter 1

Introduction



1.1 Introduction to VLSI

VLSI is an acronym for very large-scale integration, which enables integration of hundreds of millions of transistors on a small silicon chip of a few square millimeter size. This technology is solely responsible for the small sizes of heavily loaded capabilities of the electronic gadgets and gizmos of today, ranging from any type of mobile phone to smart consumer infotainment product, to smart servers, to household electronic devices. The dominant VLSI technology being CMOS technology follows the famous Moore's law "the number of transistors in a chip doubles every 18 months" which is proven correct since it was stated in 1965. However, this growth in density of transistors posed and continues to pose innumerable challenges to the designers who are required to upgrade their skills constantly to address them.

1.2 Application Areas of SOC

System on chip (SOC) has become an indispensable part of many products in almost all domains. There are SOC's being deployed traditionally in communications, data storage, and high-tech computing domains since VLSI days, and with high-level integration including analog, sensor technologies, low-power capabilities, and high signal processing possibilities, SOC is penetrating into domains like medical, automotive, security, and defense.

1.3 Trends in VLSI

The trends in growth of VLSI technology can be classified under the following heads:

- Complexity
- Speed of operation
- Die size
- Design methodology

1.3.1 Complexity

Since the time, transistors were invented; for over the past five decades now, physical dimension of the transistor is constantly shrinking. This has resulted in packing more and more transistors on a silicon wafer integrating more and more functionalities into the circuits. This phenomenon called scaling is still continuing. But, it is said that in the next 2 to 3 years, scaling of transistor's dimensions will reach a point where it will be so expensive that it becomes commercially not viable to scale down further. However, all these years the predicted demise of the Moore's law has been repeatedly proven wrong. Even today there are many other technologies beyond CMOS technologies, which appear promising in offering alternate solutions in continuing the everlasting thirst for more and more functionality in devices of reduced form factor. It is the scaling, however, which is responsible for the tremendous growth in computing and communication power of the processors which has changed the way we sense, process, store, display, and communicate information of any magnitude. Over the past five to six decades, chips have accommodated circuits, which are time critical to the entire system. Today's electronic gadgets house very few components and a few interface peripherals, apart from the system on chip (SOC) and unlike the large systems of yesteryears. The trend in integrating more and more circuits to form SOC was the result of advancement in allied technologies like photolithography, fully depleted wafer technologies, high K materials, 3D stacked silicon wafer technologies, etc. This was supported by the enhancements in EDA tools and enhancing algorithms, which run in them. As per Wikipedia [1], as of 2017, the largest transistor count in a commercially available single-chip processor is 19.2 billion – AMD's Ryzen-based Epyc. In other types of ICs, such as field-programmable gate arrays (FPGAs), Xilinx's Everest/Versal [2] has the largest transistor count, containing around 50 billion transistors showing the complexity of the SOC's of current days.